

WILL SMITH

<https://willsmith.nz> portfolio: live demos, source, commit history

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SUMMARY

Software engineer, full-time as an IT contractor at an Auckland MSP since February 2026, shipping production software, operating live SaaS infrastructure, and designing a self-hosted platform that replaced third-party hosting. Computer Science / Economics conjoint from the University of Auckland, behind an independently-built portfolio spanning embedded systems, full-stack web, and applied ML. Comfortable across the stack, from vSphere and Terraform infrastructure to React front ends to fixed-point arithmetic on bare-metal hardware.

WORK EXPERIENCE

IT Consultant (Contractor) — Novatec Solutions, Auckland

February 2026 – Present

An MSP for small-to-medium businesses. Primary technical contractor, full-time, working alongside the owner across software development and infrastructure.

Software development

- Built an invoicing automation system integrating Pax8 (cloud billing) and Xero (accounting) via Windmill; fuzzy-matches customers across systems and drafts invoices with LLM-assisted content cleanup; live, with final polish ongoing.
- Built a Python HTTP server exposing Microsoft 365 (mail, calendar, OneDrive, tasks) via per-user OAuth; deployed and in active use.
- Designed and built a live, multi-tenant AI agent platform end to end: per-tenant isolation, Docker/Terraform VM provisioning, secrets management, and identity/auth design.
- Operate and maintain it: root-causing and fixing faults, building its observability layer (structured audit logging + Prometheus monitoring), and running a security-incident response (credential rotation and git-history remediation after a leaked key).
- Built company and client web properties in TypeScript/React, collaborating with another developer on the company's own marketing site.

Infrastructure & hosting

- Designed and stood up a self-hosted client-hosting platform (Coolify on on-prem, UPS-backed ESXi virtual machines), migrating client sites off third-party hosting and eliminating its recurring cost.
- Built **novatec.co.nz** end to end: site, DNS configuration, and Cloudflare Tunnels.
- Migrated several existing client WordPress sites onto the new platform.
- Hands-on VM administration (ESXi/vSphere) and core networking (WAPs, switches, routers, DHCP).

PROJECTS

Live demos, source, and commit activity: <https://willsmith.nz>

PromptTech AI Sandbox — Capstone Project (Industry Client), 2025 — **graded A**

- Dockerised full-stack web application for comparing prompts and parameters across generative AI models: FastAPI/Python backend, React front end, PostgreSQL, AWS Cognito auth, deployed on AWS EC2.
- Set up the project's architecture and containerisation, wrote the FastAPI service skeleton, the initial AWS Bedrock integration and the pytest harness, worked on the Cognito auth flow's dev/prod configuration, and built the deployment pipeline (Python/Paramiko: EC2 provisioning, image transfer over SFTP, remote Docker Compose) that the team shipped on for the rest of the project.
- Industry client: PromptTech (reference below).

Recipe Web App — live at willscokbook.nz

github.com/QuickWaller/recipe-website-v2

- Full rewrite of an earlier recipe app: Astro + React islands, Drizzle ORM/Postgres, Docker Compose. Unit, end-to-end, and visual-regression tests, gated by CI.
- Self-hosted via Coolify on the home lab below, deployed over Cloudflare Tunnel; the same hosting stack built professionally at Novatec, run independently at home.

Home Lab Infrastructure — personal, private repo

- Two-node Proxmox cluster with live inter-node VM migration; zero open inbound ports (Cloudflare Tunnel for ingress, Tailscale-only admin access).
- Built a least-privilege permission system for it: Proxmox API roles and access-control lists give each automated AI agent its own restricted resource pool and scoped token, tested to allow only in-pool actions.

Hiking Data Logger / Weather Pod

github.com/QuickWaller/hiking-weather-pod

- Low-power field logger on an RP2350 MCU: I2C weather sensor and RTC, UART GPS, e-ink display, rule-based and ML weather forecasting, SD-card sync.
- Production-scale ML forecasting pipeline: LightGBM trained on ERA5-Land reanalysis against GPM satellite rainfall, ~31M training rows, deployed to the pod's firmware, not lab-only.
- Evaluated as real research: skill-scored against a climatology baseline (Brier/CRPS), sensor-noise simulation, and model bake-offs, with feature parity to the on-device C++ firmware via cffi.

ESP32 FM Synthesiser

github.com/QuickWaller/esp32-fm-synth · .../esp32-synth-fixed-point-wip

- Real-time FM audio synthesis on ESP32, streamed over Bluetooth A2DP: table-lookup oscillator, ADSR envelope, multi-voice mixing.
- Separate fixed-point rework: a from-scratch Q16.16/Q32.0 fixed-point ADSR envelope generator, working around the absence of hardware floating point.

Agent Tooling & Claude Code Harness

github.com/QuickWaller/claude-code-managed-repo-template

- Supervision harness for unattended AI coding sessions: permission requests race a local popup against a Telegram bot, first answer wins, failing closed on every error path.
- 72-assertion test suite with a subprocess harness against an isolated home directory; custom subagent definitions and a handoff workflow.

Vintage Story Game Server

github.com/QuickWaller/vintage-story-server

- Enhanced fork adding automated mod management (40+ mods), Coolify deployment, and structured logging.

TECHNICAL SKILLS

Languages	C, C++, C#, Python, Java, TypeScript/JavaScript, HTML/CSS
Frameworks	React, Node.js, Express, FastAPI, Flask, Arduino framework/PlatformIO, Godot, Unity
Cloud & infrastructure	AWS (EC2, Bedrock), Docker, Terraform, Coolify, VMware ESXi/vSphere, Proxmox VE, Cloudflare Tunnels, Tailscale, Linux administration, Git, core networking (VLANs, DHCP, routing)
Data & APIs	PostgreSQL, REST, WebSockets, OAuth
Embedded	ESP32/ESP32-CAM, RP2350, fixed-point arithmetic, real-time audio DSP, low-power design, I2C/UART peripheral integration
AI/ML	Generative AI API integration, applied ML forecasting pipelines (LightGBM, satellite/reanalysis data fusion, skill-score evaluation), MCP servers and agent tooling, machine learning fundamentals

EDUCATION

University of Auckland — Bachelor of Commerce / Bachelor of Science (Conjoint), Computer Science & Economics
2021 – 2026, graduated

- **COMPSCI 399 Capstone: Computer Science** — A (industry-client project for PromptTech, see Projects)
- **COMPSCI 361 Machine Learning** — A
- **COMPSCI 235 Software Development Methodologies** — A-

Kristin School — NCEA Levels 1–3 Graduated 2020

- Dux 2019 and 2020 · Kristin Cup for Excellence in Mathematics, 2020 · Achieved with Excellence at all levels.

ADDITIONAL EXPERIENCE

Waiter / Line Cook — Long Bay Surf Club November 2025 – February 2026

Volunteer, Breakfast Service — Auckland City Mission January 2025 – August 2025

- Supported Auckland's homeless and vulnerable community at a soup-kitchen-style breakfast service; a high-stress environment requiring reliability and composure.

Calculus & Statistics Tutor — Self-managed February 2024 – July 2025

- Ran a one-on-one NCEA tutoring business: found students, scheduled and priced sessions, wrote lesson plans and materials.

Commis Chef — Beau 2022 – 2024

- High-end French cuisine; trained new team members; managed ordering and stock.

Earlier roles — Dough Boys Bakery, Michael Hill Jewellers, KFC 2018 – 2021

REFERENCES

Amir Mohammadi — AI Adoption and Development Lead, PromptTech — amir@promptech.ai